



Building Enterprise-Ready AI Agents with Microsoft Foundry with Guardrails

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Principal Architect



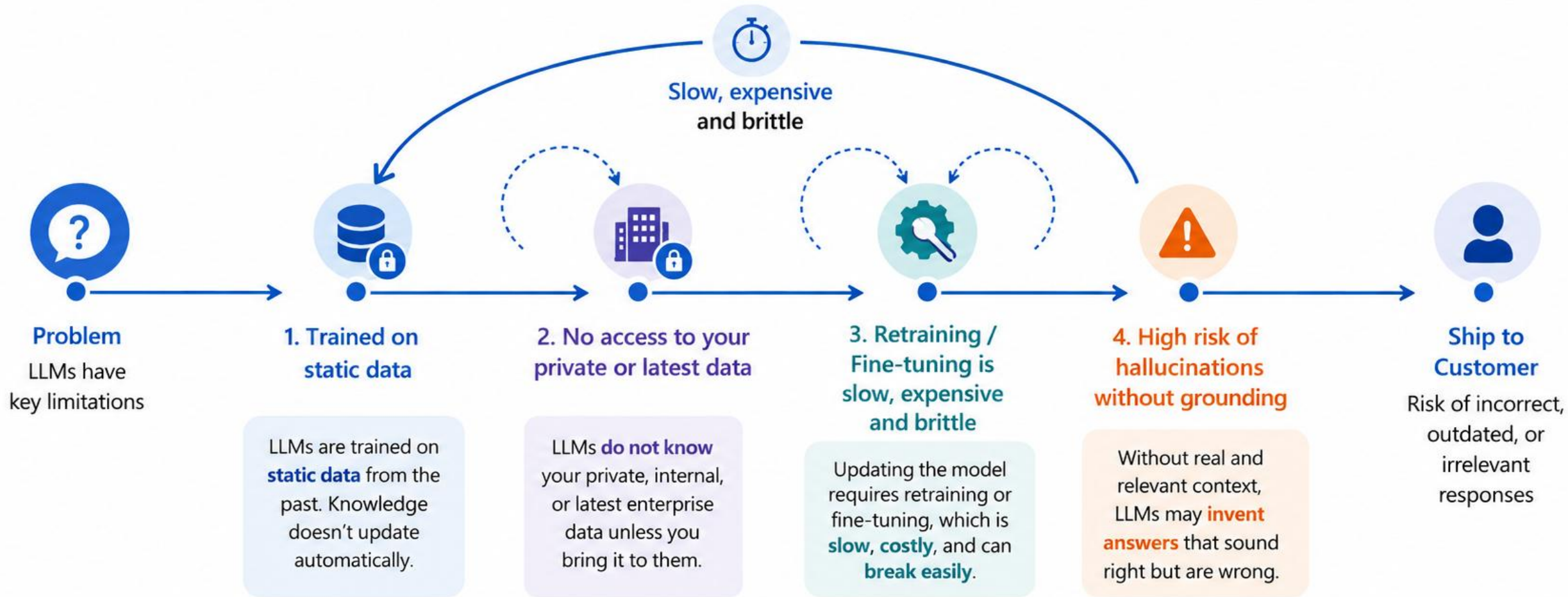
Agenda

-
- RAG 101
 - Knowledge Bases in Azure AI Search
 - Microsoft Foundry and Knowledge Bases
 - Observability and Guardrails
 - Q&A

Introduction to RAG



Problems with LLMs



Why it matters:

Without addressing these issues, LLMs can produce outdated, irrelevant, or incorrect answers—creating risk for your business and users.



Microsoft
Copilot

Ground LLMs with your enterprise data using RAG to deliver accurate, secure, and up-to-date responses.

How RAG Resolves LLM Limitations

RAG – Retrieval-Augmented Generation



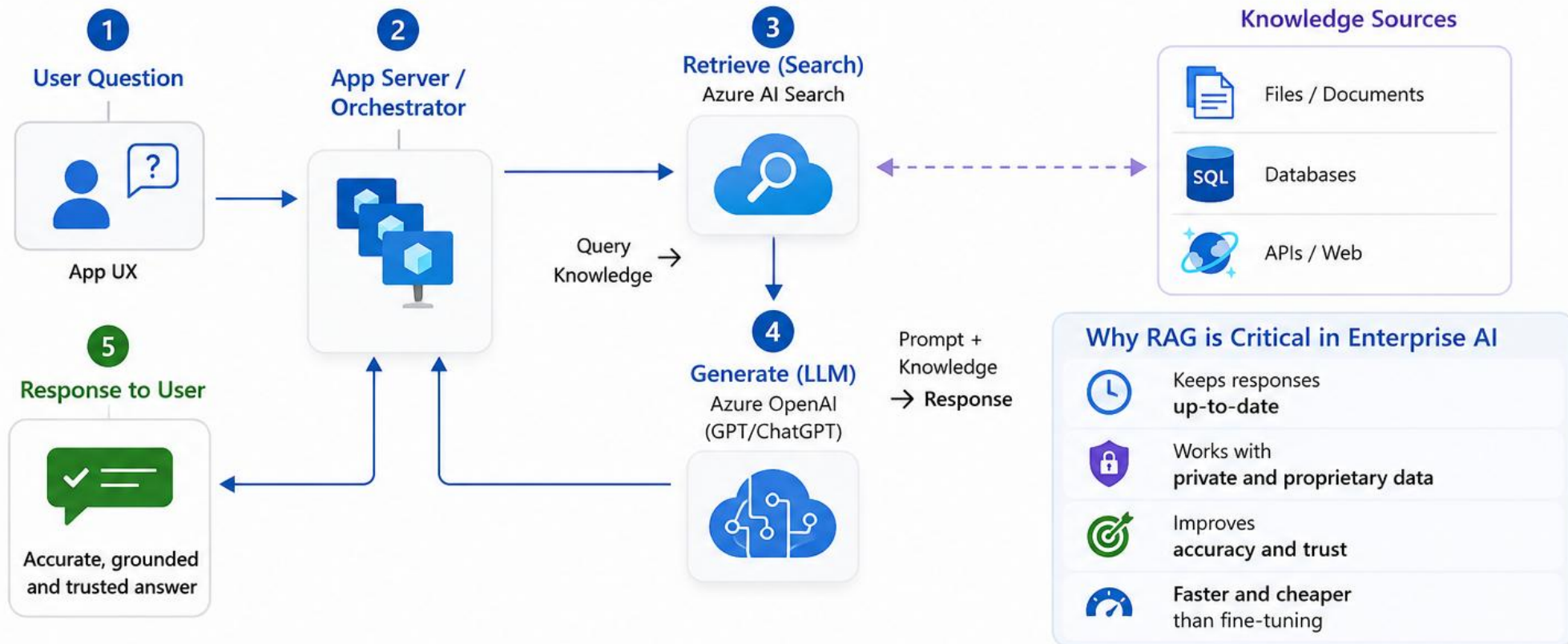
Architecture that grounds LLM responses in external knowledge



Combines search + prompt augmentation + generation



Solves hallucination, freshness, and private-data limitations of LLMs



The Result:

RAG grounds LLMs in your enterprise knowledge, delivering accurate, current, and trustworthy answers.



Microsoft
Copilot

Smarter AI. Your data.
Better outcomes.

How RAG Resolves LLM Limitations

RAG – Retrieval-Augmented Generation



Up-to-date
Uses latest data



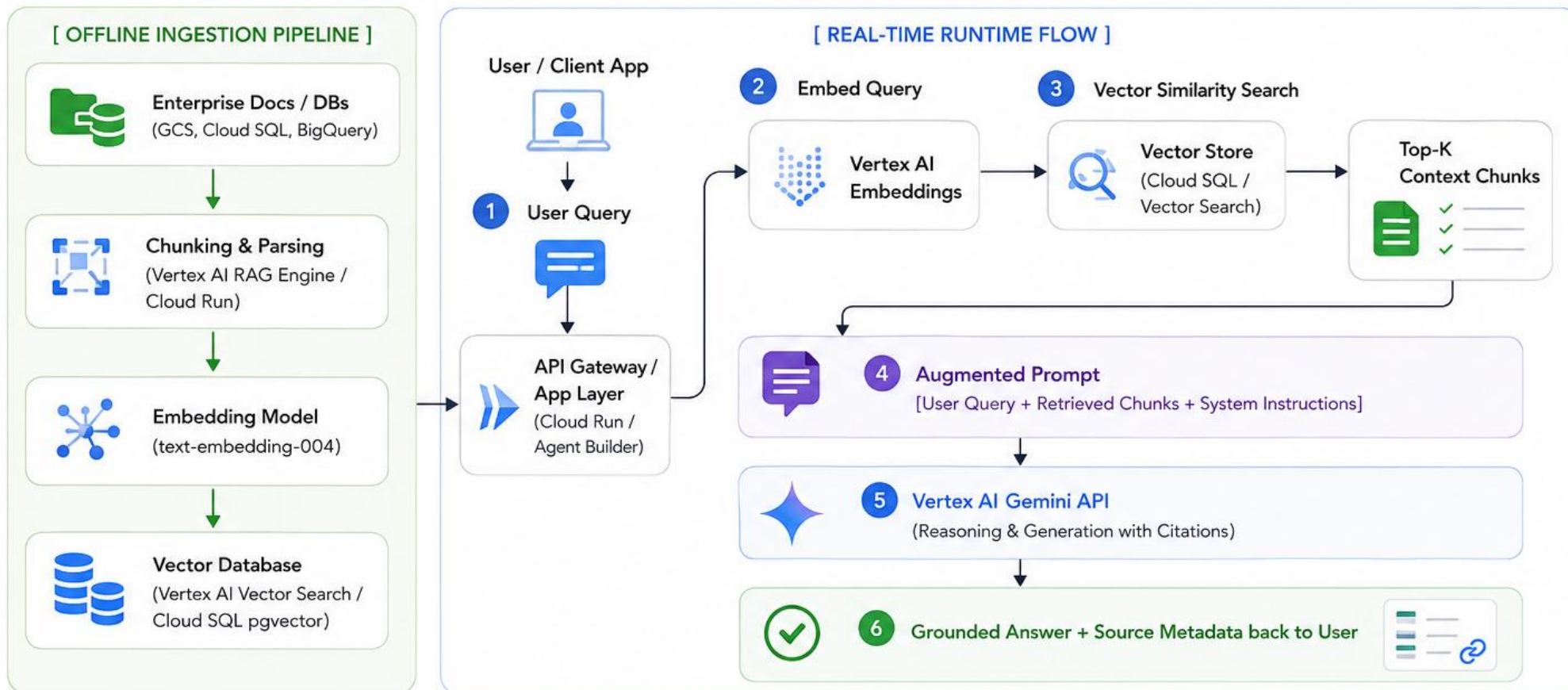
Private & Secure
Works with your
enterprise data



Accurate & Trusted
Reduces hallucinations
with grounding



Efficient
Faster and cheaper
than fine-tuning



The Result

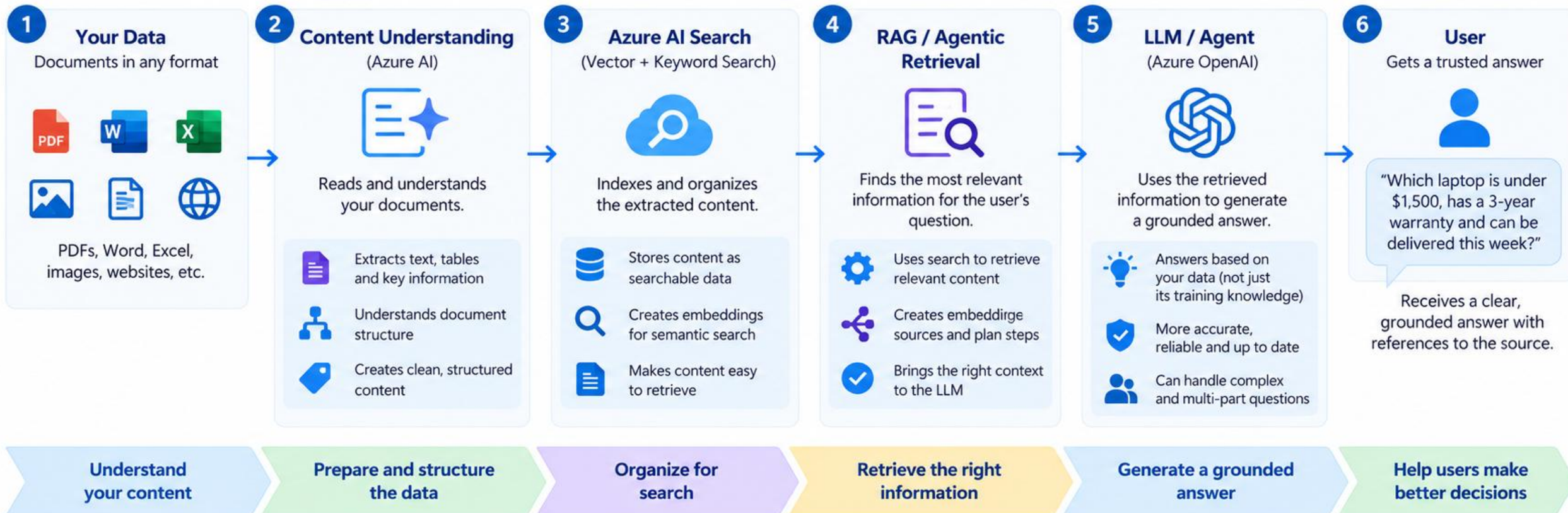
RAG grounds LLMs in your enterprise knowledge, delivering accurate, current, and trustworthy answers.



Smarter AI. Your data.
Better outcomes.

How RAG Connects to Content Understanding in Azure AI

Content Understanding prepares your data. RAG finds and uses that data to give better answers.



 **Key takeaway:** Content Understanding makes your data easy to search. RAG uses that data to give better, more reliable answers.



File Search Tool

Grounded enterprise knowledge retrieval

Search enterprise
documents

Automatic
retrieval

Built-in agent
integration

Grounded,
cited responses



Azure AI Search

State-of-the-art retrieval

Rigorously tested
retrieval

Streamlined
data pipeline

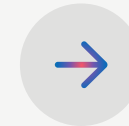
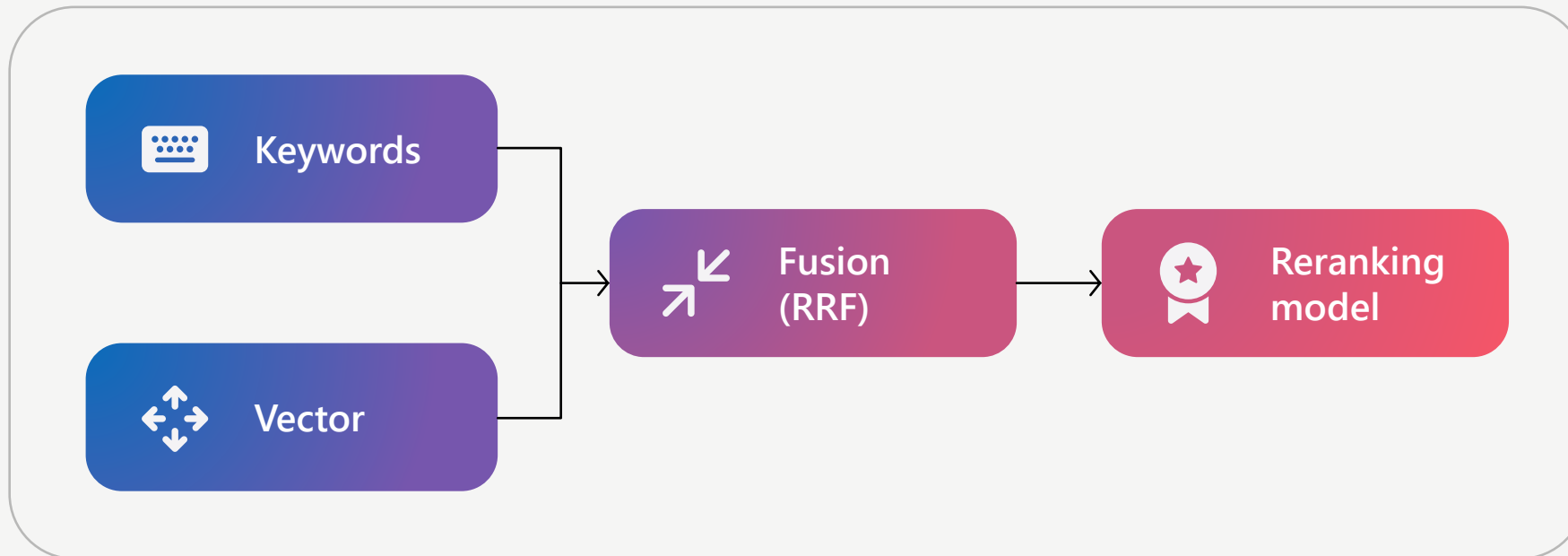
Developer
integrations

Enterprise-ready
foundation

Hybrid search: Optimal retrieval strategy

Complete search stacks do better:

- Hybrid retrieval (keywords + vectors) > pure-vector or keyword
- Hybrid + Reranking > Hybrid



Let's break those down...

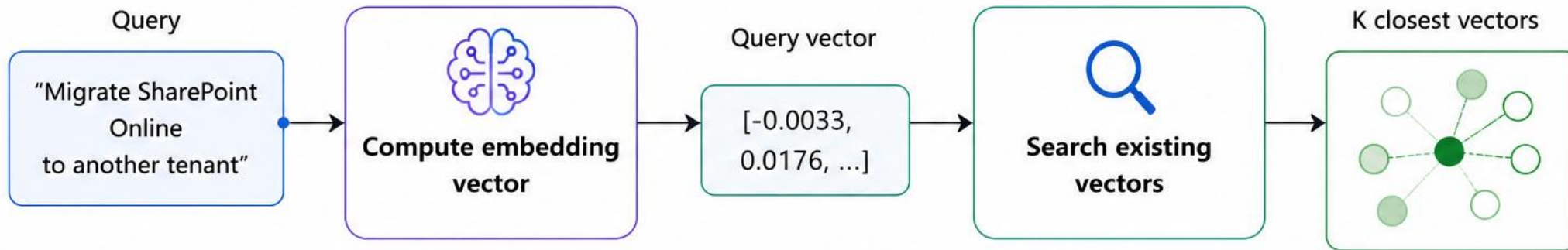
1. Vector Search



Best for: conceptually similar matches

Vector search

- 1 Encode queries as embeddings (lists of [floating point](#) numbers)
- 2 Find semantically similar documents by comparing embeddings



Example search results

Document	Cosine Similarity
Cross-tenant SharePoint migration guide	0.91
Move Microsoft 365 content between organizations	0.87
Transfer SharePoint sites to a new Microsoft 365 tenant	0.84



Azure AI Search can use the **HNSW** approximation algorithm (*Hierarchical Navigable Small World*) for more efficient search across massive vector indexes.



Simple meaning: Vector Search finds documents with similar meaning, even when they don't contain exactly the same words.

2. Keyword Search



Best for: exact phrase/keyword matches

Keyword search

Documents are stored in an **inverted index**, which can be queried efficiently.

Documents

ID	Name	Description
335	SharePoint Migration Guide	Migrate SharePoint Online sites to another tenant
334	Microsoft 365 Migration	Move Microsoft 365 workloads between organizations

Inverted Index

Term	Document List
migrate	335
SharePoint	335
Online	335
tenant	335, 334

Then the user searches:



Search results

ID	Document	Relevance
335	SharePoint Migration Guide	0.92
334	Microsoft 365 Migration	0.61
312	SharePoint Tenant Planning	0.48



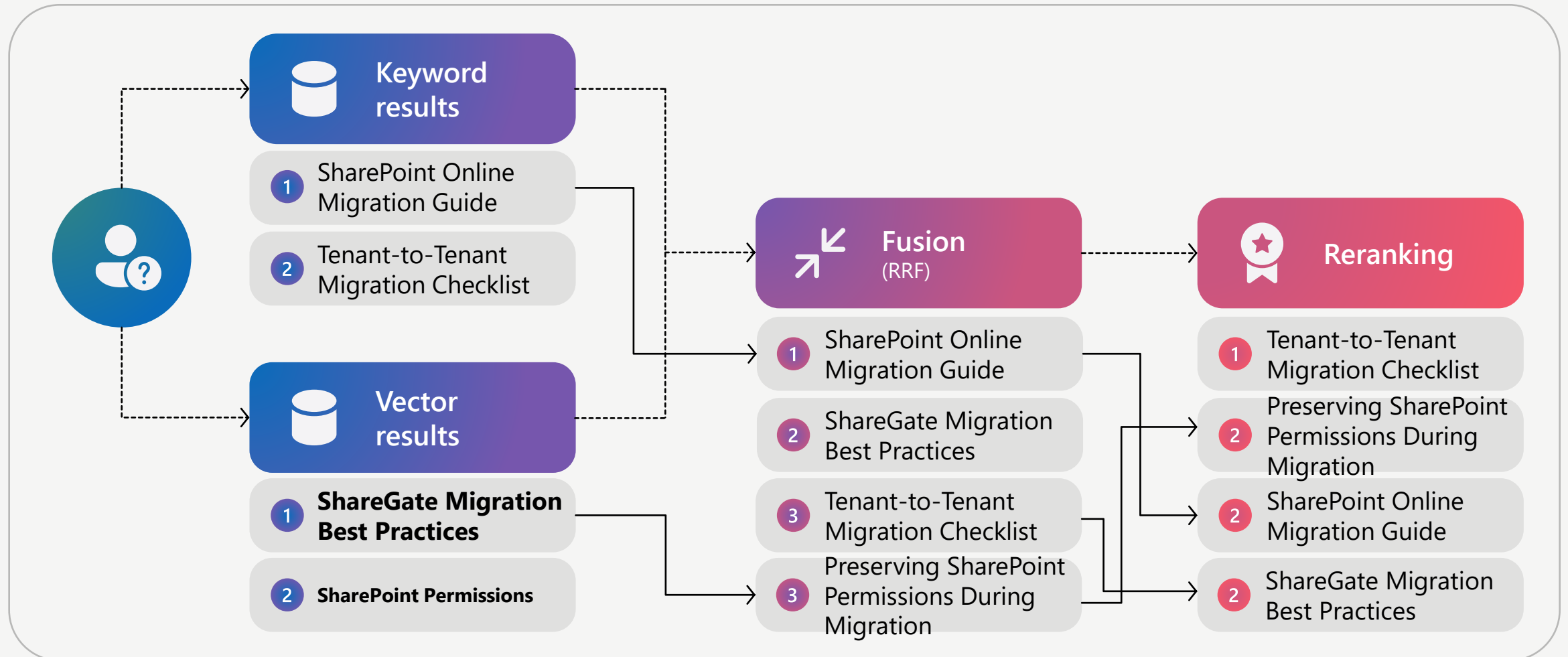
Azure AI Search uses the **BM25** algorithm for ranking full-text search results.



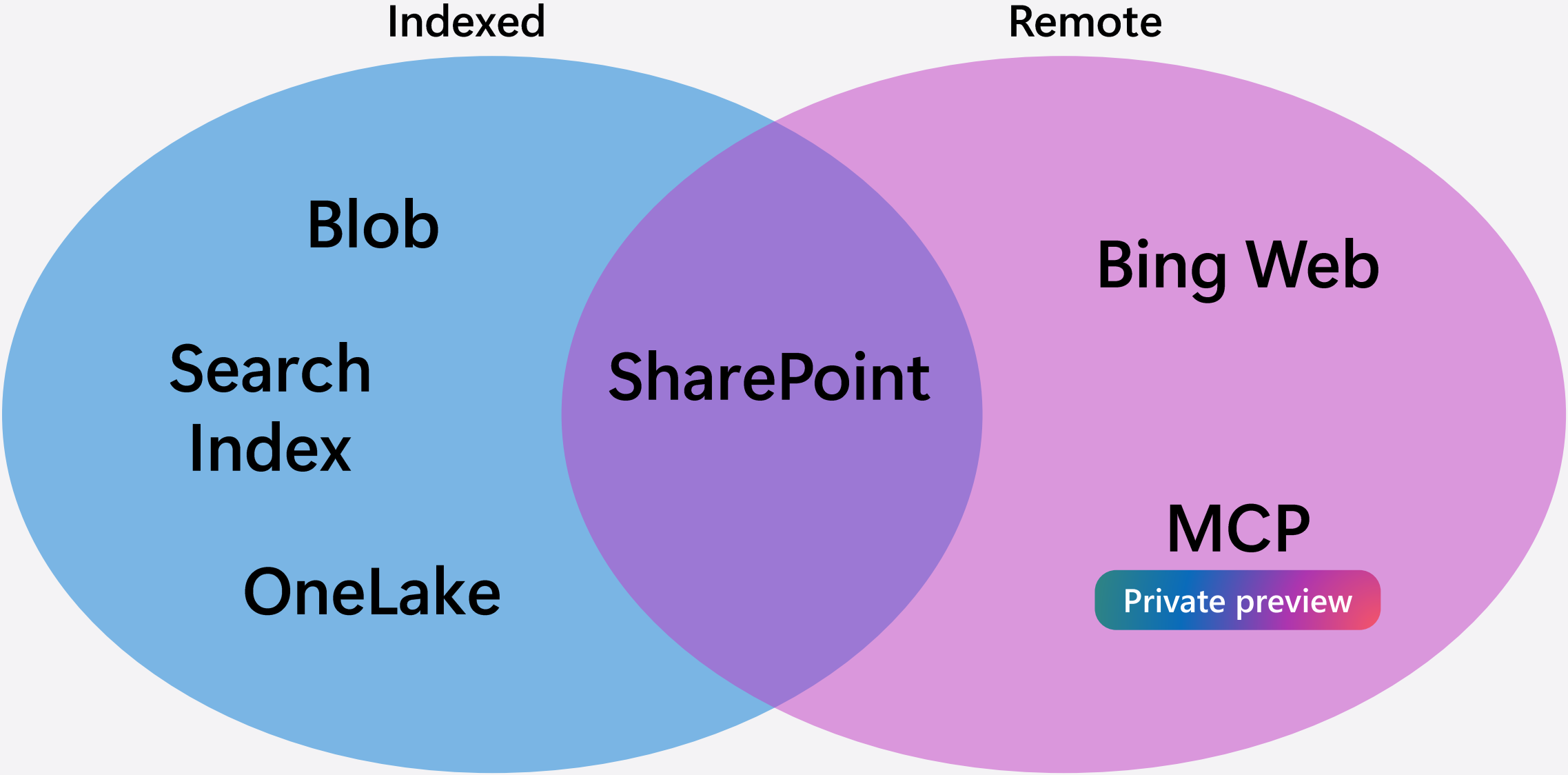
Simple meaning: Keyword Search finds documents containing the same or closely matching words you searched for.

A complete hybrid search flow

Question: "Migrate SP2023 to SharePoint "



Indexed and Remote Knowledge Sources



Demo 1 – Agentic RAG

Microsoft Foundry implementation of RAG with Azure AI Search



What Makes Agentic Retrieval Different?

It's not just about finding information — it's about planning, using multiple sources, and giving better answers.



Normal Search / Classic Retrieval

Finds and returns information



Returns a list of relevant documents or text
(ranked results)



Follows a single search path
(one query, one set of results)



Does not understand your full conversation
(limited context)



You usually configure and manage it manually
(no planning or decision making)

VS.



Agentic Retrieval

Plans, finds and uses the right information



Plans the retrieval steps
(decides what to search for)



Can search across multiple sources
(files, databases, websites, etc.)



Uses your conversation context
(understands follow-up questions)



Can run multiple searches in parallel
(faster and better results)



Better suited for complex questions
(gives more accurate and useful answers)



Key idea: Normal search finds content. Agentic retrieval plans how to find, use and combine the right information to give you a better answer.

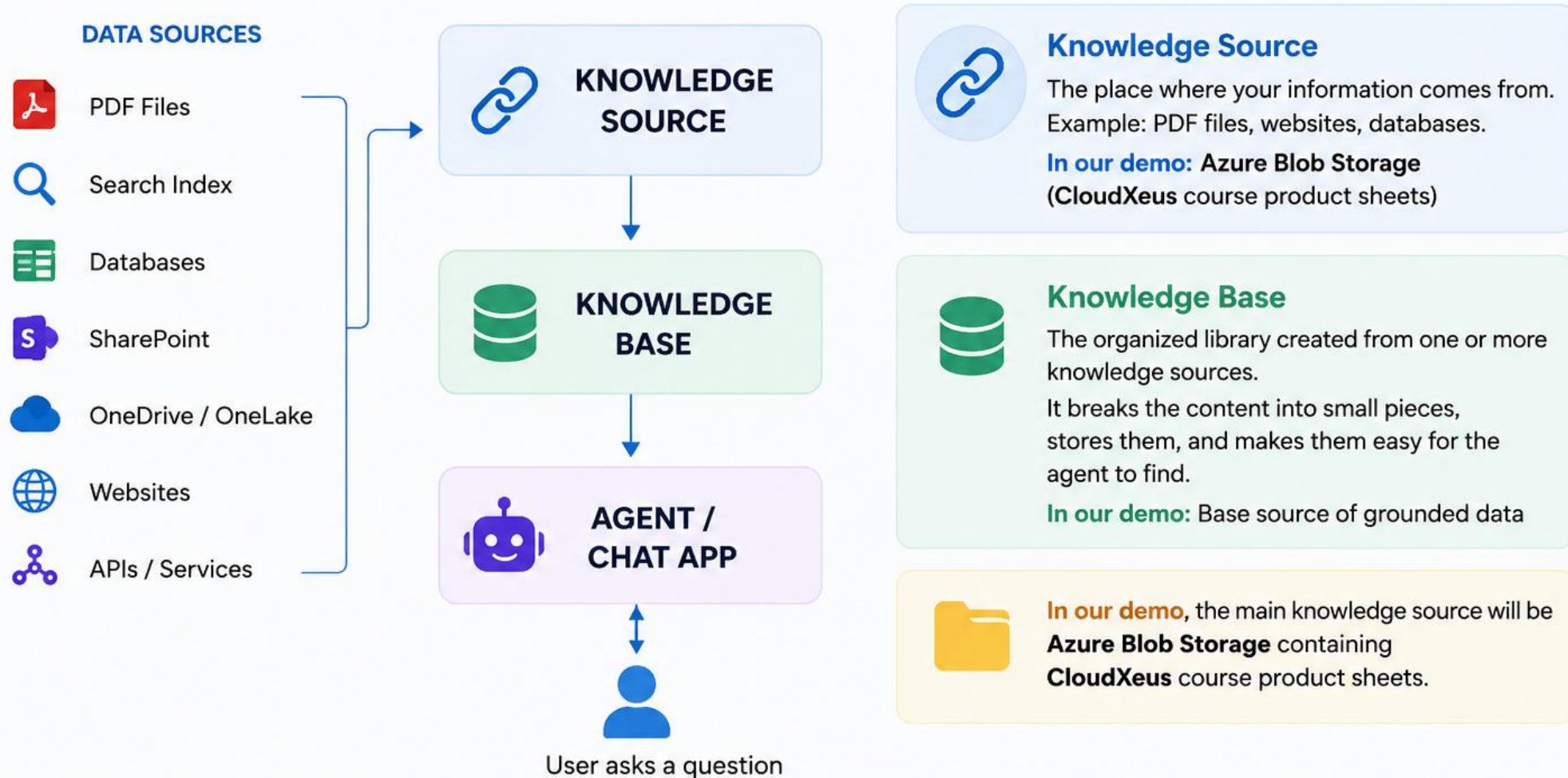


Why this matters: In our demo, the agent uses agentic retrieval to answer CloudXeus product questions more intelligently than a basic search.

Microsoft Azure

RAG in Simpler Words

How your agent finds the right information and gives better answers



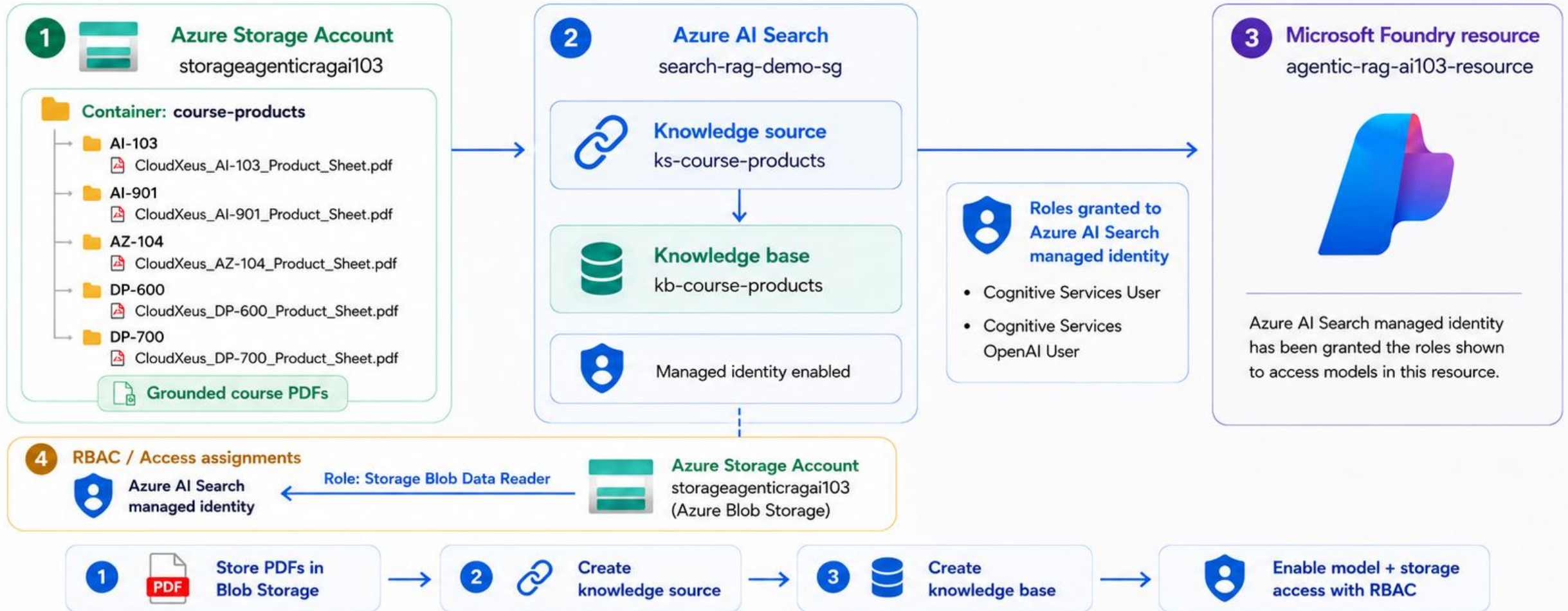
Why this matters:

Agents need grounded data (real and verified information) before they can answer reliably and help users with confidence.



Agentic RAG Setup Overview

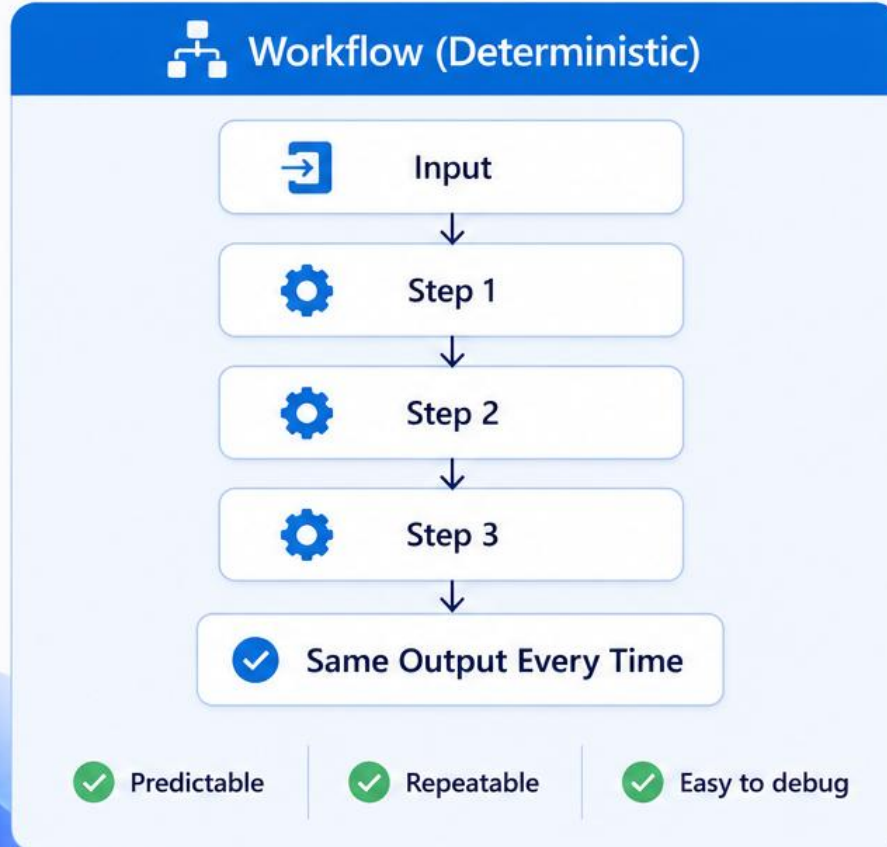
What we have configured so far for CloudXeus agentic retrieval



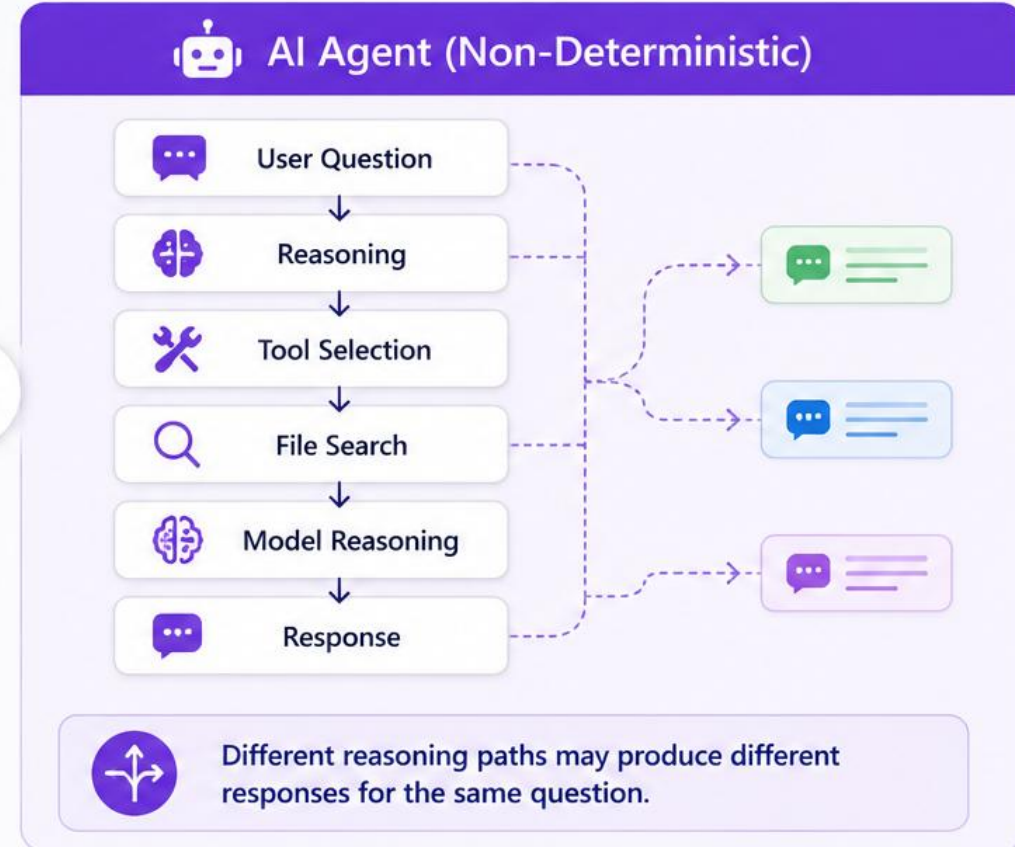
Why this matters: Azure AI Search can now read the CloudXeus PDFs, use Microsoft Foundry models for enrichment and retrieval, and build the grounded foundation for our agent.

Why AI Agents Need Observability

AI agents are non-deterministic. Observability gives us the visibility to understand, debug, and improve their behavior.



VS.



Observability brings clarity to non-deterministic behavior so you can build trustworthy, production-ready AI agents.



Inspect



Measure



Debug

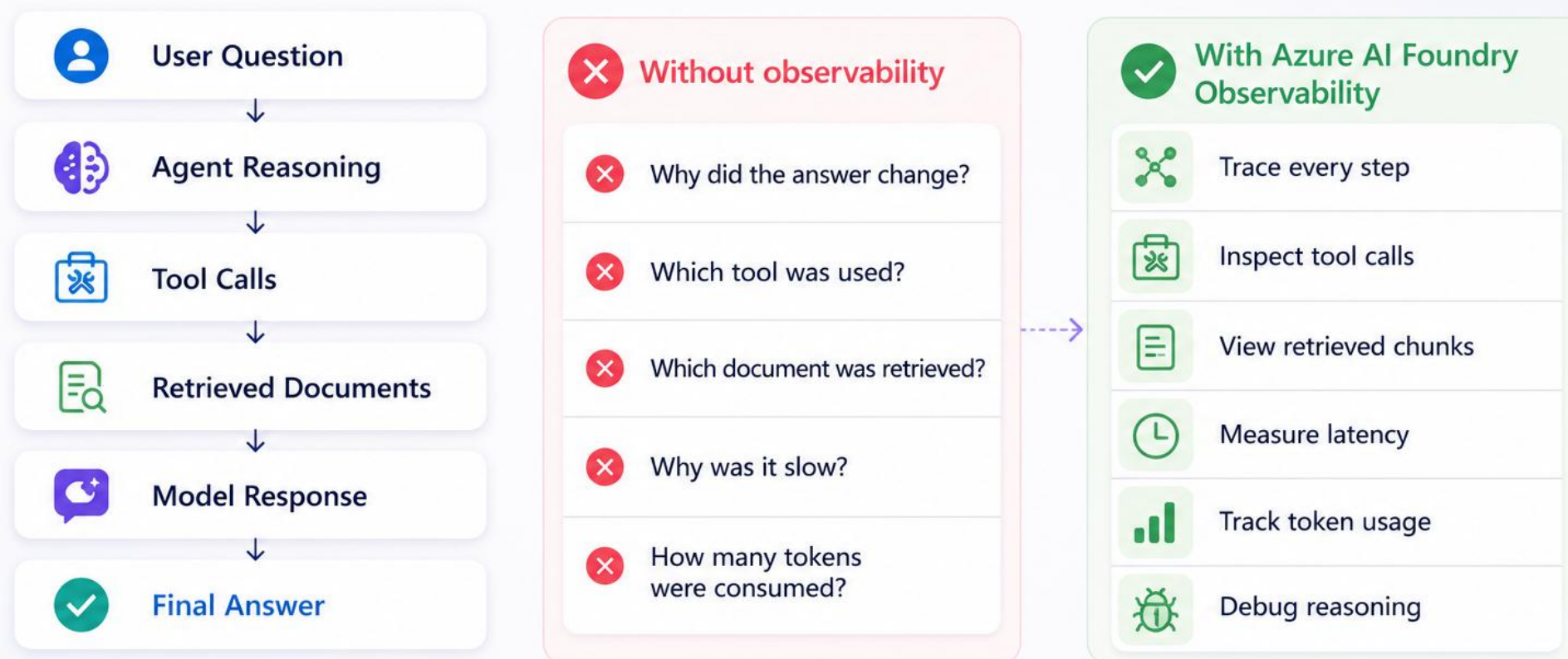


Govern



Improve

Observability Makes AI Agents Explainable



Full visibility. Better insights. Confident AI.

Azure AI Foundry Observability gives you end-to-end visibility to build, operate, and trust your AI agents in production.



Trust



Govern



Improve



Optimize

AI Observability in Azure AI Foundry

Trace, understand, and improve every step of your AI agent's journey.

Azure AI Foundry

Contoso Project

Search

Project home

Observability

Overview

Traces

Metrics

Evaluations

Alerts

Dashboards

Build and customize

Model catalog

Playgrounds

Prompt flow

AI Services

My assets

Models + endpoints

Data + indexes

Tools

AI Foundry > Contoso Project > Traces > Trace details

← Trace: 8f3c2e2a-7b6d-4fa7-9c11-1d9e2b8c3f45 Completed

Conversation ID

User

Start time

Total duration

Total tokens

c6d8b7a1-6f3e-4e80-9a21-0f7b9a6c8d2e

Aditi Anand

May 14, 2025 10:42:18 AM

12.45s

2,134

Trace view | Timeline | Metadata | Logs

Name	Duration	Status
Conversation Complete user interaction	12.45s	✓ Completed
Invoke Agent Agent starts processing	1.21s	✓ Completed
Execute Tool File Search (SharePoint Index)	2.73s	✓ Completed
Retrieved Documents 15 chunks retrieved	1.18s	✓ Completed
Chat gpt-4o reasoning and response generation	6.82s	✓ Completed
Final Response Answer returned to user	0.51s	✓ Completed

Trace Component	Purpose
Conversation	Complete user interaction
Invoke Agent	Agent starts processing
Execute Tool	File Search, API, SQL, Function
Chat	GPT reasoning and response generation
Duration	Time taken for each step
Input + Output	Data sent and received
Metadata	Model, tokens, IDs, status, configuration



Observability answers the question:
"Why did my AI agent produce this response?"



Full Visibility



Faster Debugging



Greater Reliability



Lower Cost



Enterprise Trust

Grounding

1.  **“Grounding ensures AI responses are based on trusted enterprise data instead of the model's memory.**

- Examples:
- SharePoint Online
- Azure AI Search
- File Search
- SQL Database
- PDFs
- Microsoft Graph

Grounding vs Guardrails

Two essential pillars for reliable, responsible, and trustworthy AI.

 Grounding	 Guardrails
Where does the AI get its facts?	What is the AI allowed to do, say, or generate?
 Uses trusted enterprise knowledge	 Enforces safety, security, and compliance
 Reduces hallucinations	 Prevents harmful or unsafe responses
 Retrieves relevant information	 Applies organizational policies
 Improves answer accuracy	 Controls AI behavior



Grounding tells the AI what to know.

It connects the model to trusted data so answers are accurate and relevant.



Guardrails tell the AI how to behave.

It sets boundaries to ensure safe, compliant, and responsible outputs.



Attacks on Agents:



Prompt injection



Jailbreak attempts



Abusive or unsafe user input



Requests for protected or disallowed content



Microsoft treats guardrails as a first-class, system-level feature in Foundry.



In Foundry, guardrails are implemented through **content filtering and risk controls**, powered by **Azure AI Content Safety**.



They work by **scanning both**:

User Input
(Prompts)



Azure AI Content Safety



Detects and blocks harmful, unsafe, or policy-violating content in real time

Model Output
(Responses)



Guardrails help protect your users, your data, and your brand—at every step.



Protect users



Protect data



Ensure compliance



Build trust

Demo 2 – Agent's Guardrails Observability

Microsoft Foundry Traces and Group



Conclusion



Resources and next steps



Get Started with Azure AI Search
aka.ms/AISearch-new



Foundry IQ Public Preview
aka.ms/FoundryIQ



GitHub Sample – Quickstart
aka.ms/AISearch-ar-pyn



GitHub Sample – Agent Service Integration
aka.ms/AISearch-ar-agent



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Sign up for the MCP Private Preview
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